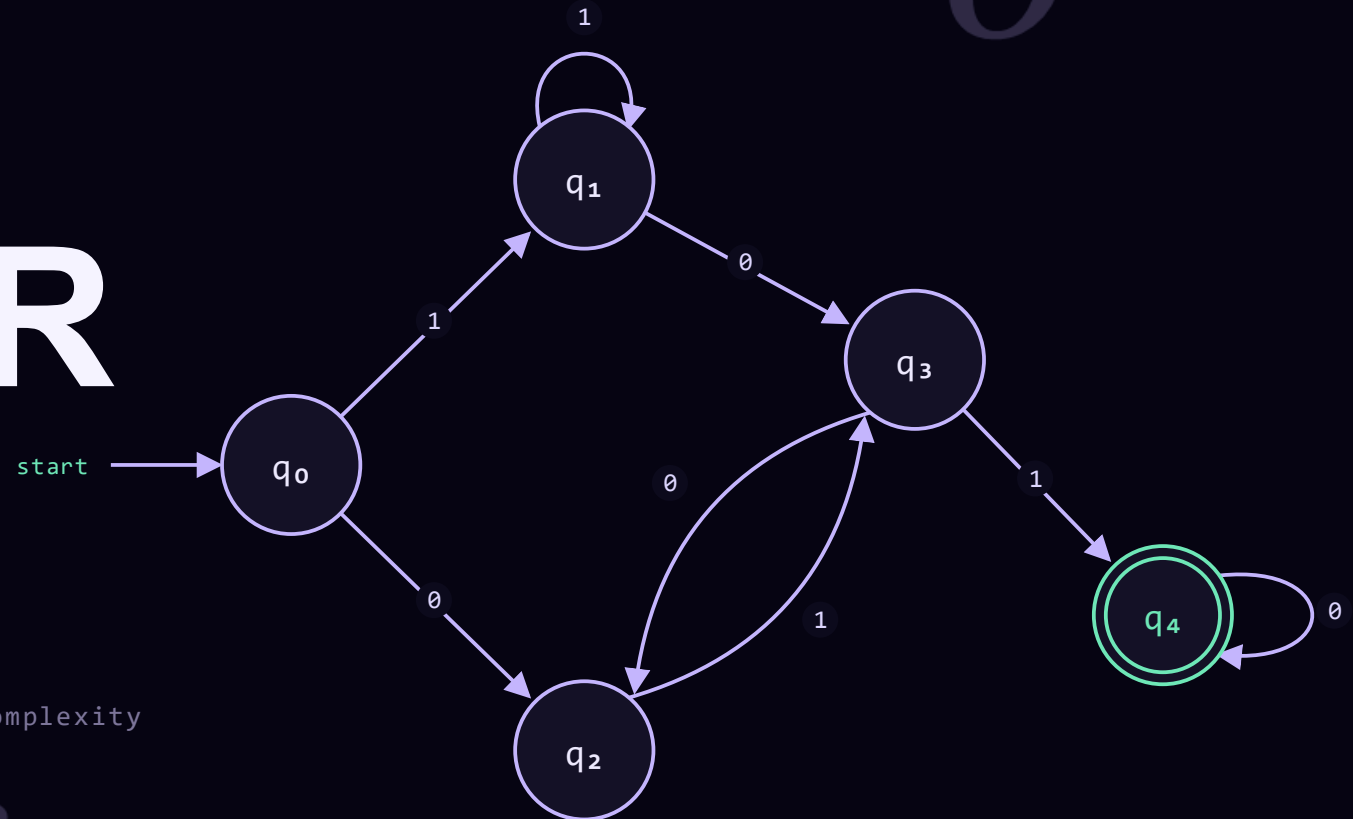


CSCI 338 • FALL 2026

COMPUTER THEORY

What can be computed — and what can't.

automata • grammars • Turing machines • decidability • complexity



finite automaton over $\Sigma = \{0, 1\}$ • double circle = accept

TURING MACHINE • TAPE



Instructor details

- Adiesha Liyanage
- Email:
a.liyanaralalage@montana.edu
- Office hours:
 - Monday 12:05 PM 02:00 PM
 - Tuesday 10:05AM 11:00 AM
- Office: GH 260E (2nd floor)



Activity

- Let's go around the class—please tell us your name, where you're from (City/State), and your major and something fun you did during the summer break.

TA details

- **Name:** Braedan Sopp
- **Day(s):** TBD
- **Time:**
- **Location:** Student Success Center
- **Email:**

Let's look at several problems

True or False?

- Finding shortest path in a distance-weighted graph can be done in polynomial time

True or False?

- Finding shortest path in a distance-weighted graph can be done in polynomial time

True

True or False?

Finding the longest simple path (no repeated vertices) in a given undirected graph can be done in polynomial time.

True or False?

Finding the longest simple path (no repeated vertices) in a given undirected graph can be done in polynomial time.

Highly unlikely

True or False?

- Can you build a program that will answer the question: “Will this input code terminate eventually on a given input?”

True or False?

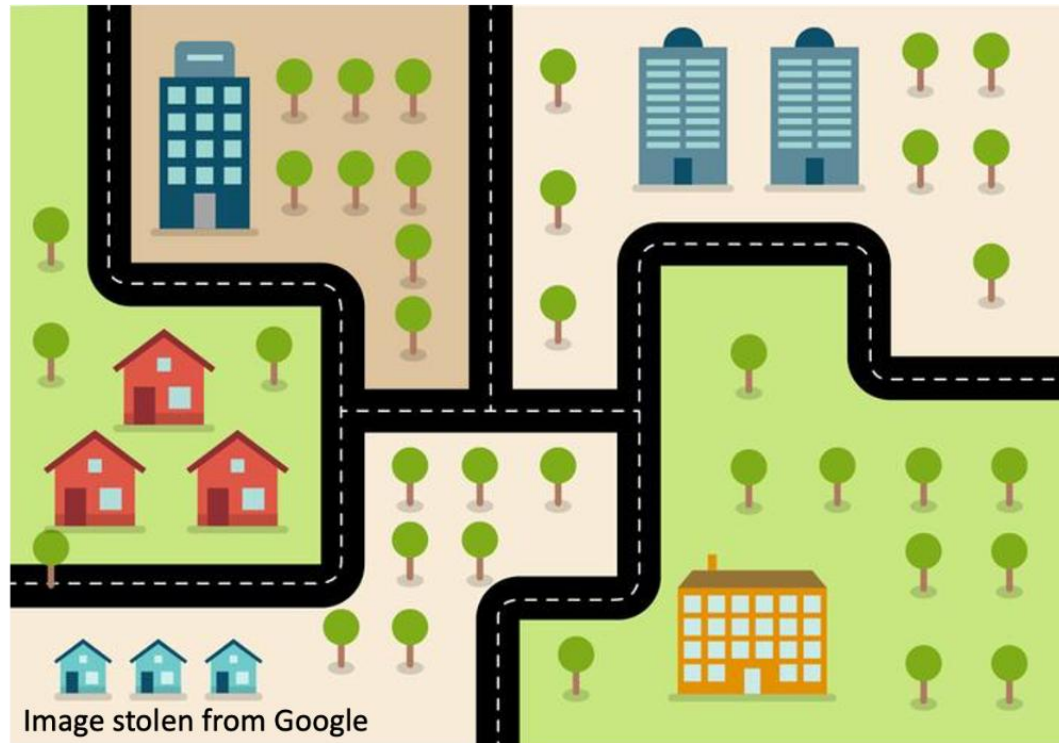
- Can you build a program that will answer the question: “Will this input code terminate eventually on a given input?”

**Impossible regardless of the computer
(Unsolvable)**

Mathematical model

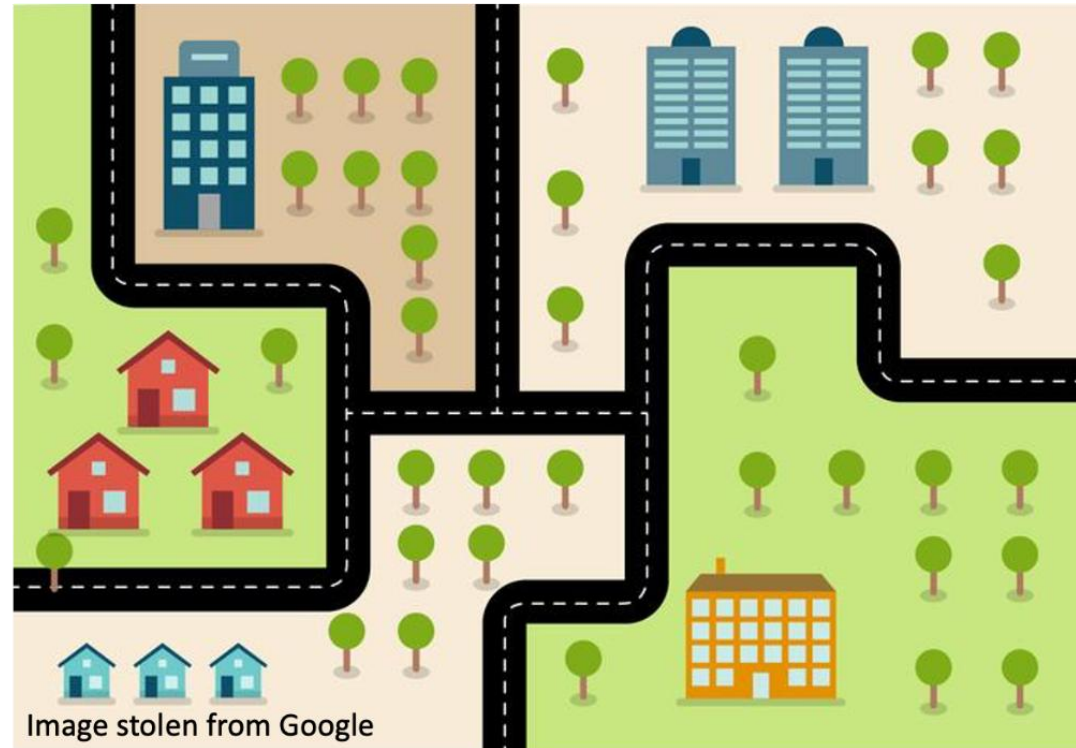
- A rigorous mathematical formulation of reality.
- Used to make predictions.

Can we represent a road network using a mathematical model?



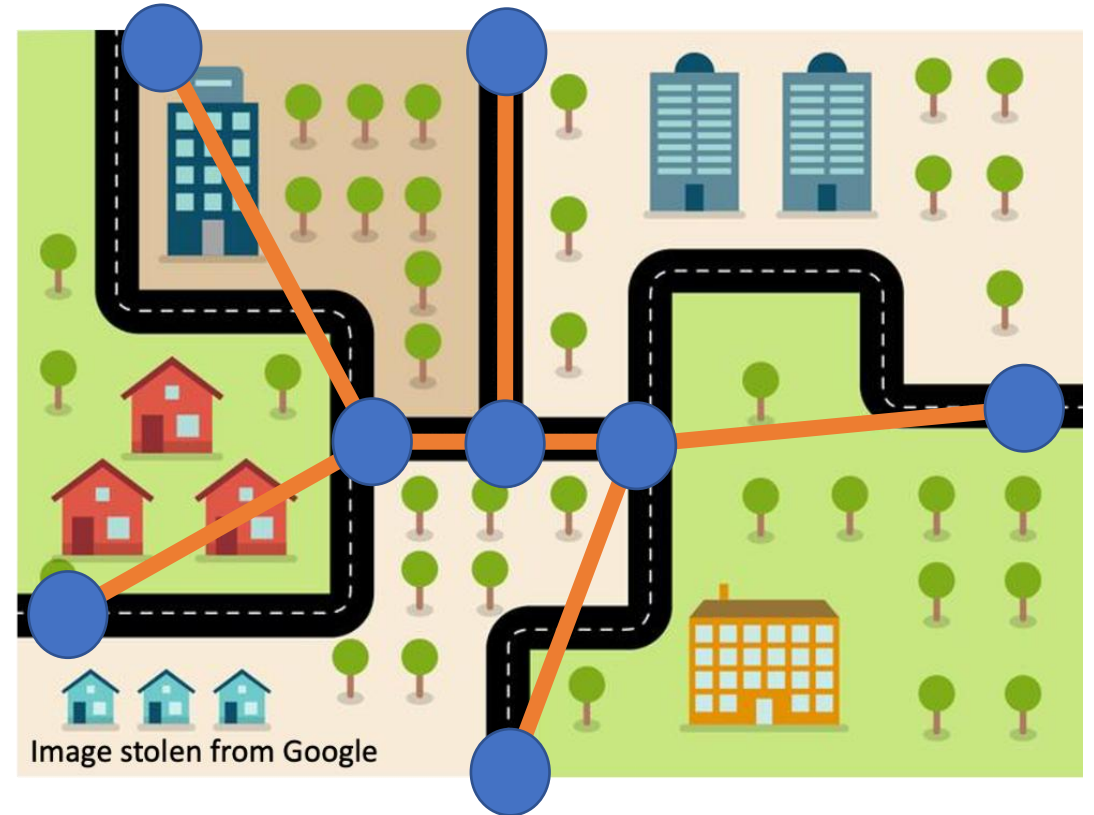
Can we represent a road network using a mathematical model?

- Graph?
 - Nodes/edges?
 - What should we use node for?
 - What should we use edges for?
 - Node for junctions?
 - Edges for road segments connecting junctions?

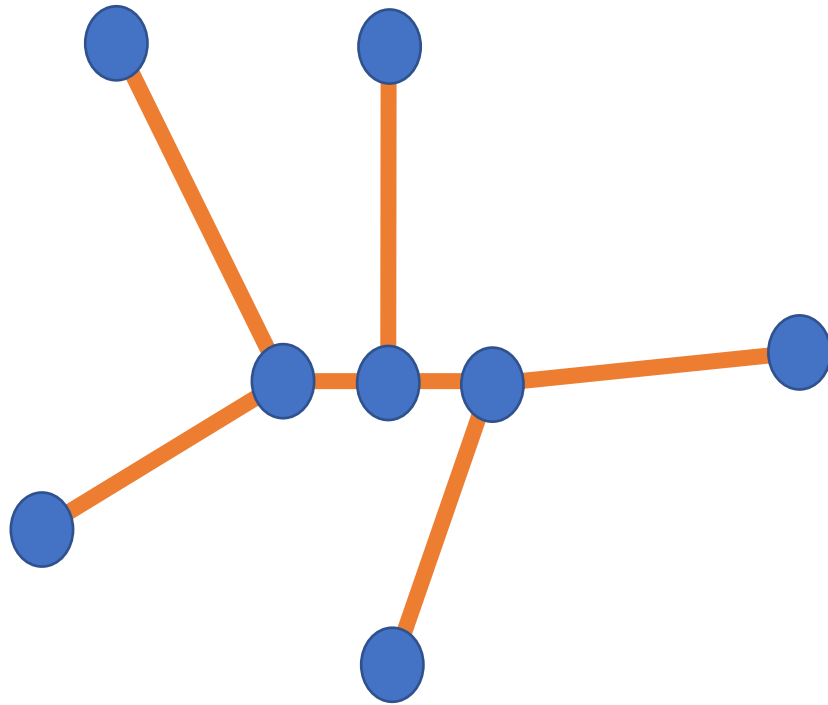


Can we represent a road network using a mathematical model?

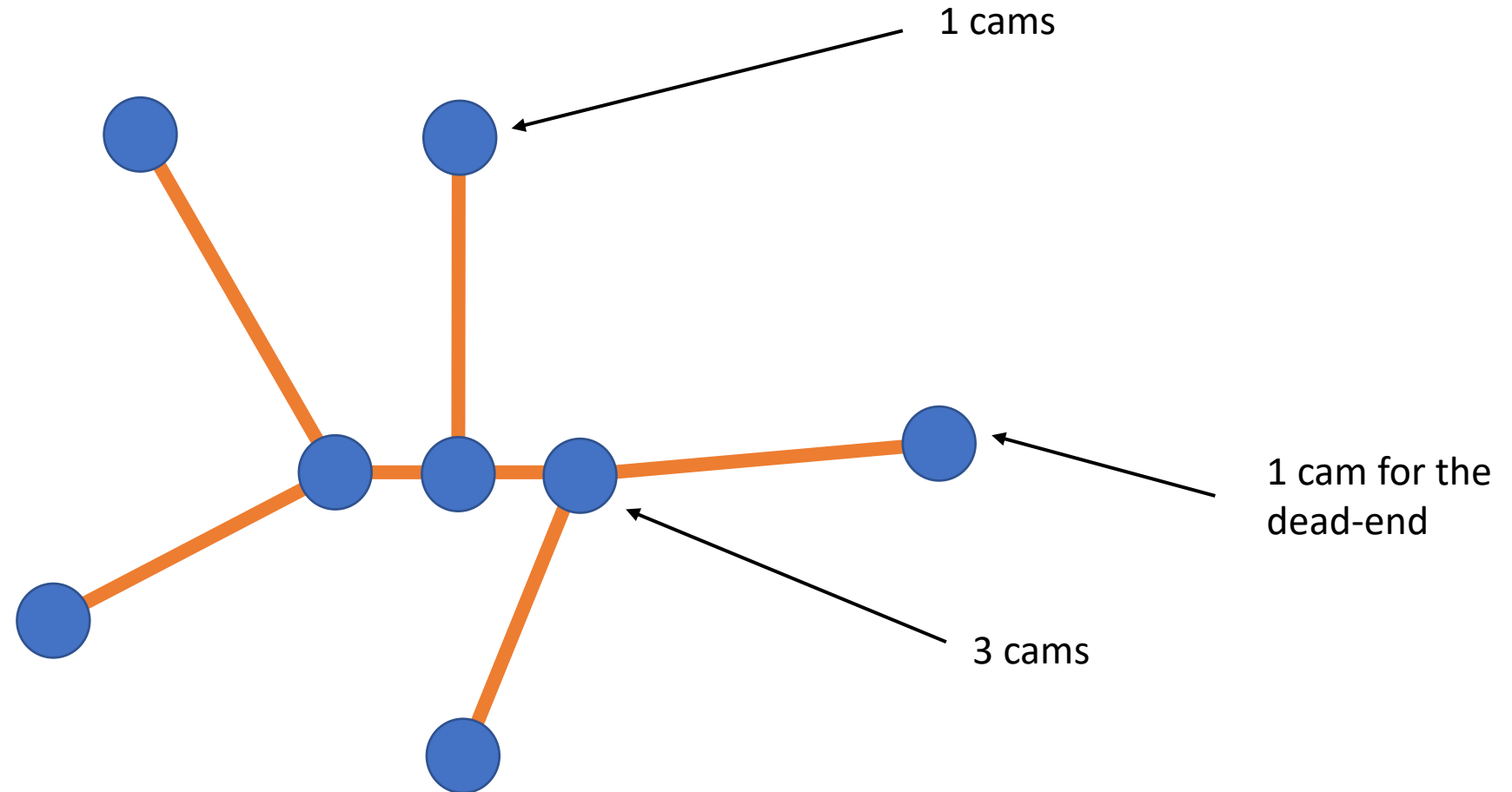
- Graph?
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 - Edges for road segments connecting junctions?



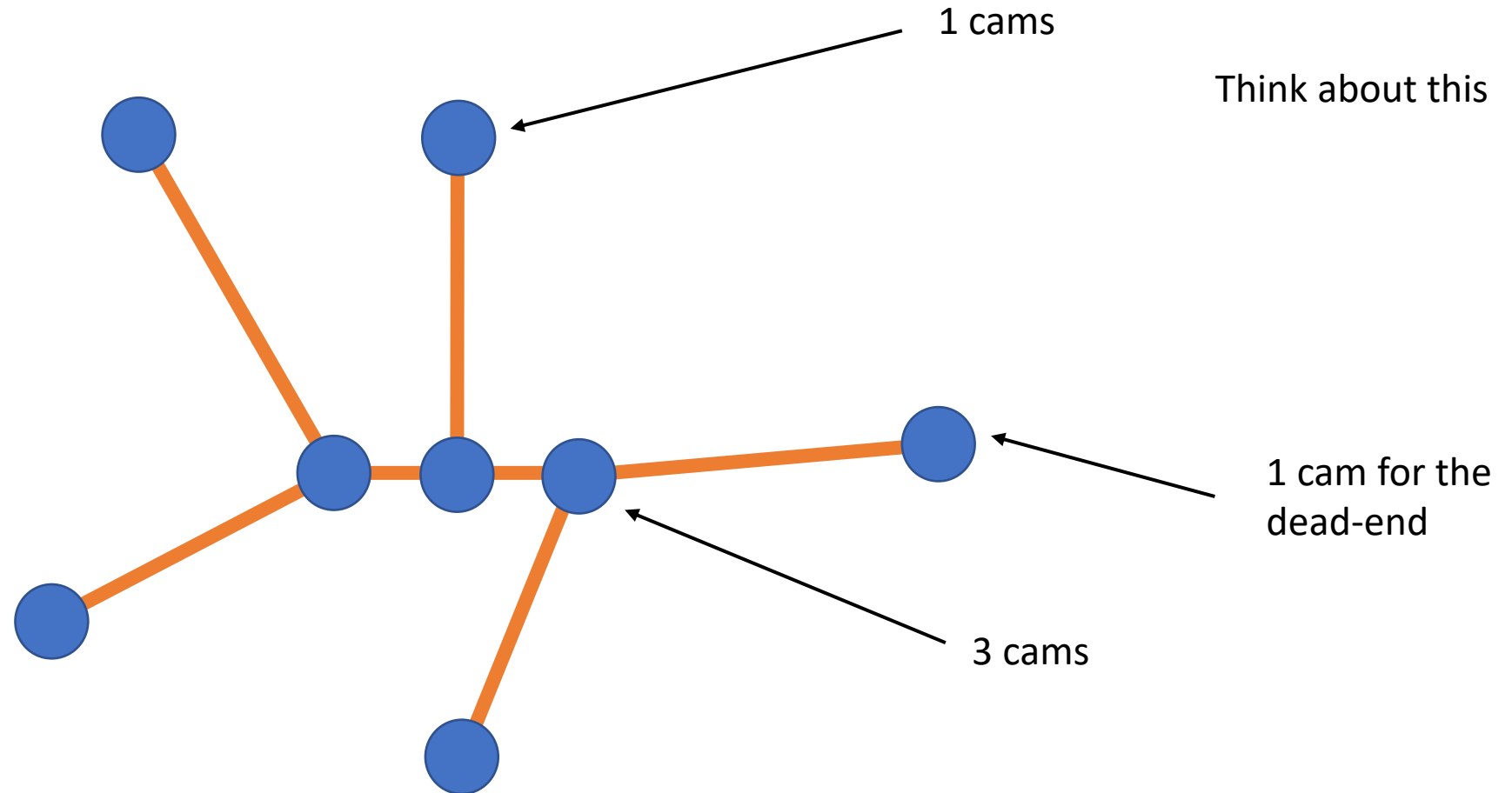
Now, let's see whether we can answer questions using this model



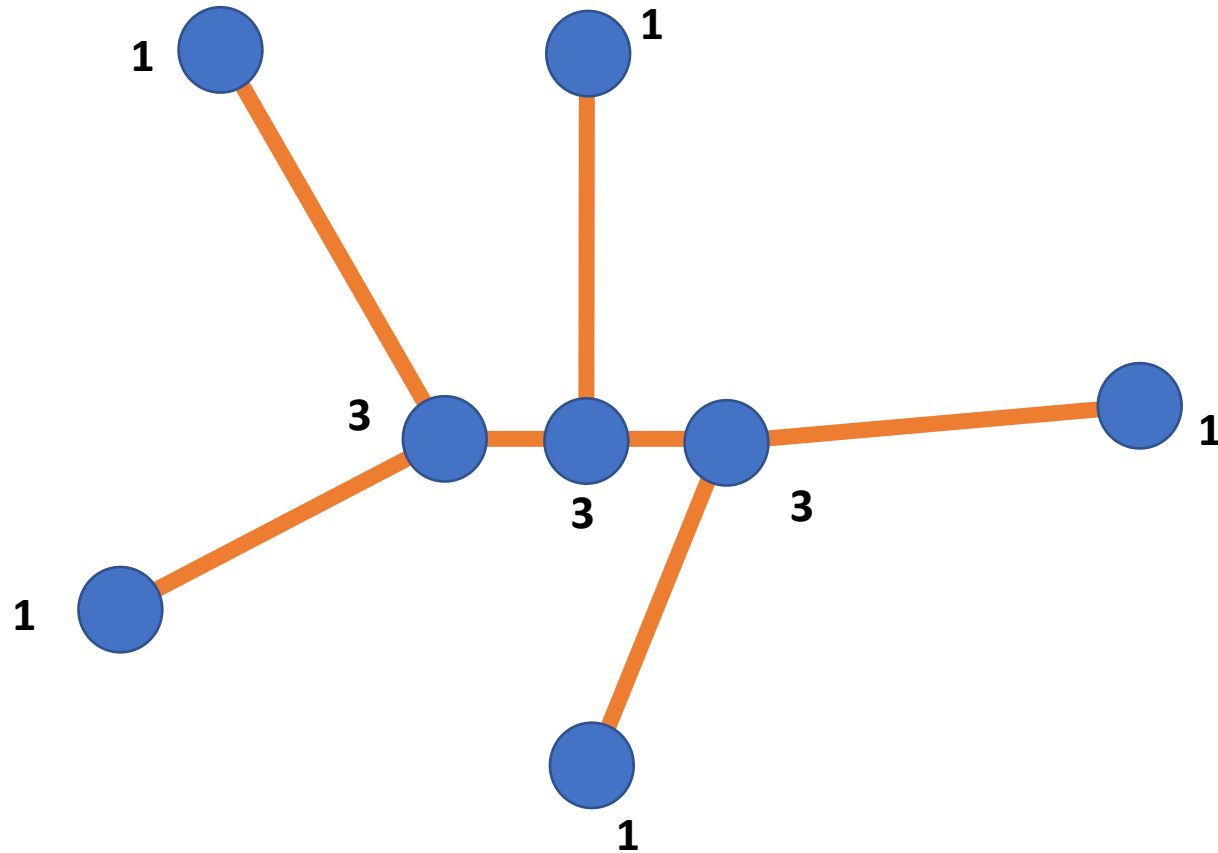
Each intersection requires a camera to monitor each road segment



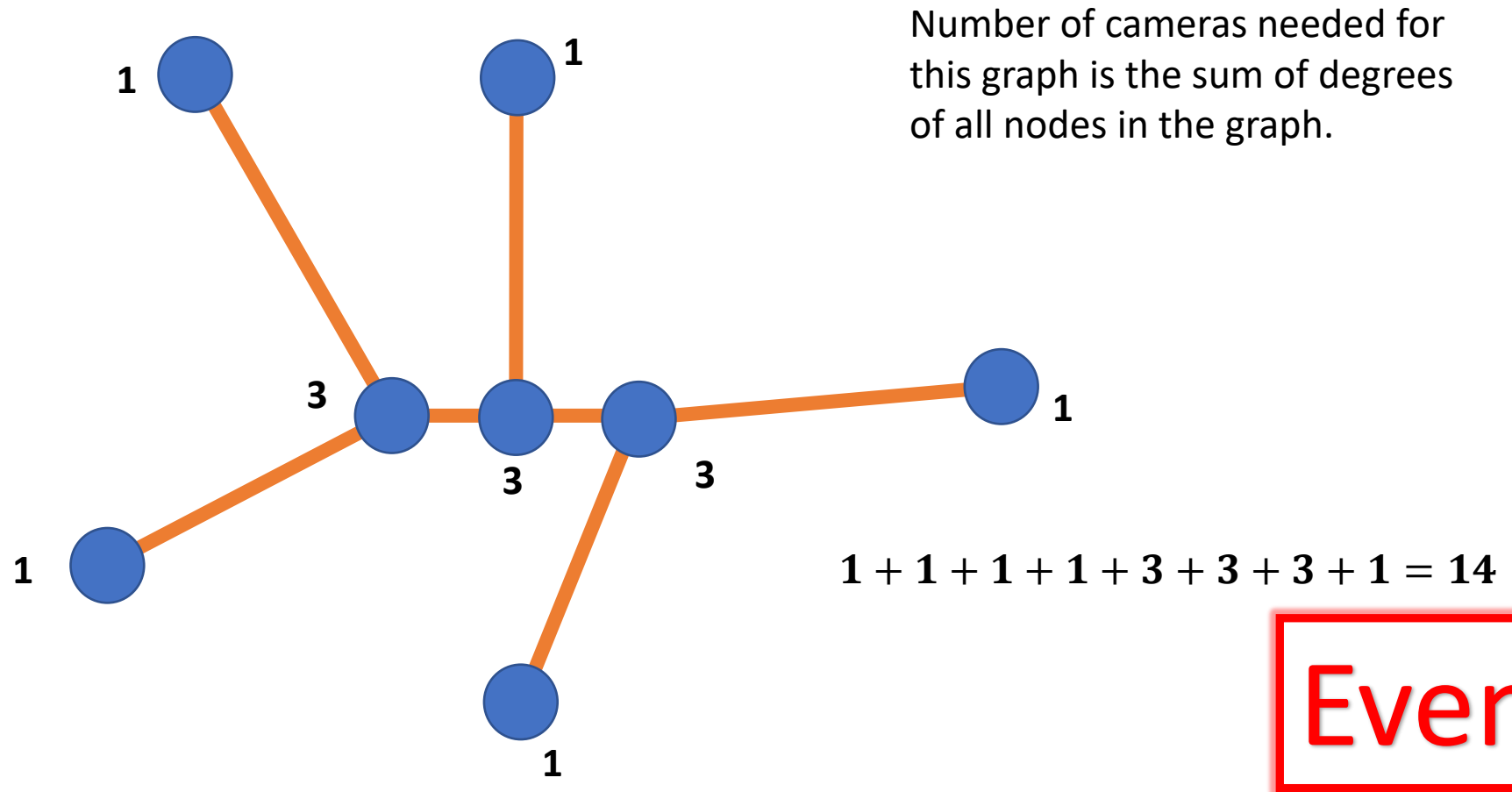
Can we build a road network such that required number of cameras is odd?



Can we build a road network such that required number of cameras is odd?



Can we build a road network such that required number of cameras is odd?



Can you answer this question for any road network?

- Suppose all road networks can be represented as a graph
- What can you say about the sum of the degrees of any graph?
- Each edge is considered twice when we consider the sum of degrees.
- Sum of degrees will always be even.
- Therefore, the answer to the question “Can we build a road network such that required number of cameras is odd?” is **NO**

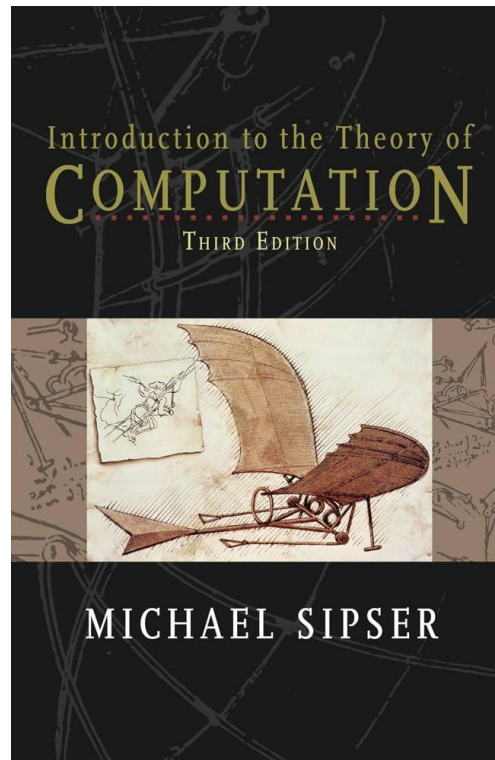
Summary

- We considered an ill-defined thing.
- We built a formal model to represent it.
- Found limitations of the model, which translated to the limitations of the thing.
- This is basically what we are going to do in this class.

- We will learn fundamental capabilities and limitations of computers.
- We will essentially learn about 3 things.
 - Automata theory
 - Automata theory deals with the definitions and properties of mathematical models of computation.
 - Ex: finite automaton, push down automaton, Turing machines.
 - Formal definitions of computers and what each types of computers (models) can do and cannot do.
 - Complexity theory
 - We will try to classify problems based on how hard the problem is to solve.
 - What makes some problems computationally hard and others easy?
 - Computability theory
 - Study of whether a particular problem is solvable or not using a computer.
 - We will try to classify problems by those that are solvable and those that are not.

Textbook

- Introduction to the theory of computation by Michael Sipser.
- This is a theory course; you **MUST** read the book.



Communication

- Post your questions on MSU CSCI Discord channel.
 - CSCI-338-Computer-Science-Theory (CATEGORY)
 - <https://discord.gg/dh7kDU4S>
- I will post the material on canvas.
- I will use GitHub page for updating the schedule.
 - <https://github.com/adiesha/CSCI338Fall2026>

Pre-requisites

- CSCI 232 and CSCI 246
- If you have forgotten what you learned in Discrete Structures class, then go through the notes of Discrete Structures.
 - <https://github.com/adiesha/CSCI246Fall2024>
 - This is a link to my notes in one of the previous semester's discrete structure class.
- It is important that you understand the basics in Discrete Structures before taking CSCI 338.
- I will do a recap of the basics in the first few lectures. (This is a quick overview; I will not take lot of time to explain)

Evaluation and grading

Grading	Weight of each category
Pop up Quizzes	4%
Assignments	42%
In class tests	30%
Final exam	20%
Attendance	4%
Total	100%

Pop up Quizzes

- These are in class pop up quizzes.
- They will be lightly graded.

Attendance

- 4% of your grades would be attendance.
- If you have 80% of the attendance, then you will get full grade for your attendance score.
- We will use iClicker to record attendance.
- This is a free service.
- Use the following link to register for the course.
- <https://join.iclicker.com/RWPQ>
- Register one device for iClicker attendance.
- At start of each class, you must use your device to record attendance.
- If you encounter problems with registration talk with UIT.



Attendance

- Please do not come to me at the end of the semester and tell me that you could not get the iClicker working.
- Resolve these issues in the beginning of the semester.
- The attendance server is run automatically during the class time, this would include holidays, I will remove these days manually from the server so do not worry about them.
- During the tests you do not have to record attendance.

Evaluation and grading: Tests

- 3 Tests + Final
- The final is optional. If you are happy with the grades of the 3 in-class tests, then the final test grade will be the average (in percentage) of the 3 in-class tests multiplied by 20. If you are not happy with the average of your in-class test, then you could choose to take the optional final.
- **The optional final cannot be taken early.**
- Note that quizzes are not an attendance counting, however, they will be graded lightly.
- At the end of the semester, final grades will be determined based on your overall performance on assignments, quizzes, tests and the final exam.

Accommodations

- If you have a documented disability and need or might need accommodations, please reach out to the Office of Disability Services as soon as possible and then talk to me.

Collaboration policy

- For your homework assignments, you may discuss the problem with your peers.
- However, you are not allowed to copy proofs from your peers or use generative AI to complete your assignments.
- If you use resources from the internet to solve problems, you **MUST** cite them.

Assignments and Latex

- Assignments MUST be completed using LaTeX and submitted as a PDF. There will be no exceptions to this rule—grading handwritten proofs is extremely difficult, so I must enforce this for the sake of the TA's sanity.
- You can create a free [Overleaf](#) account, which is a free platform for editing and compiling complex LaTeX documents online. This is very similar to Google Docs but with LaTeX.
- Basic [Overleaf latex](#) tutorials.
- If you do not know the LaTeX code for a specific mathematical symbol, you can use the following [website](#): <https://detexify.kirelabs.org/classify.html> Draw the symbol that you want; it will provide you the necessary code for that symbol and the packages that you have to import.
- Using LaTeX for mathematical writing is infinitely easier than using Word or handwriting the solutions once you get the hang of it.